# OFDM-DQPSK Audio Transceiver for Acoustic Communication

Aadarsh Ramachandran (EE23B001), Manoj NH (EE23B044)
Indian Institute of Technology Madras

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***Abstract*—This project describes the design and evaluation of a complete audio transceiver that transmits binary data acoustically using speakers and microphones as the physical channel. The system combines Orthogonal Frequency Division Multiplexing (OFDM) with Differential Quadrature Phase Shift Keying (DQPSK) to achieve robustness against the multipath reflections, ambient noise, and hardware-induced phase drift that are characteristic of indoor acoustic channels. A Zadoff–Chu sequence is used for frame synchronization and channel estimation, and two fixed pilot subcarriers track residual Carrier Frequency Offset (CFO) across the packet. The implemented transceiver was evaluated on a live acoustic recording and achieved a Bit Error Rate (BER) of approximately 1.39% at a data rate of approximately 7.2 kbps, using only commodity laptop hardware.**

## I. Introduction

Digital communication systems encode binary information into analog waveforms for transmission through a physical medium. The primary challenge with using Audio-band acoustic communication, using loudspeakers and microphones as transducers, is that the room introduces multipath echoes with delay spreads on the order of milliseconds, the SNR varies with distance and background noise, and consumer audio hardware often introduces non-negligible clock frequency offset between transmitter and receiver.

OFDM is the modulation framework of choice for combating multipath fading. By spreading data across many narrow orthogonal subcarriers and inserting a Cyclic Prefix (CP) longer than the channel delay spread, OFDM converts a frequency-selective fading channel into a set of independent flat-fading sub-channels, each of which can be equalized by a single complex multiply. DQPSK is chosen as the per-subcarrier modulation because it encodes information in *phase differences* between successive OFDM symbols rather than in absolute phases, making it inherently tolerant to residual phase errors that a coherent scheme such as QPSK or QAM would find catastrophic.

The primary figures of merit used to evaluate the system are:

- **Data Rate**: the number of information bits successfully transferred per second.
- **Bit Error Rate (BER)**: the fraction of decoded bits that differ from the transmitted bits.

## II. System Parameters

All design choices were driven by the constraints of the acoustic channel and standard 48 kHz consumer audio hardware. Table I lists the key parameters extracted directly from the implementation.

TABLE I
SYSTEM PARAMETERS

| **Parameter** | **Value** |
|---|---|
| Sampling frequency $F_s$ | 48,000 Hz |
| OFDM symbol duration $T_{\text{sym}}$ | 20 ms (960 samples) |
| Cyclic prefix duration $T_{\text{CP}}$ | 5 ms (240 samples) |
| Total symbol period | 25 ms (1,200 samples) |
| Number of data subcarriers | 90 |
| Number of pilot subcarriers | 2 |
| Subcarrier spacing | 200 Hz |
| Lowest subcarrier frequency | 1,000 Hz |
| Highest subcarrier frequency | 19,400 Hz |
| Bits per subcarrier per symbol | 2 (DQPSK) |
| Bits per OFDM symbol | 180 |
| ZC preamble sequence length | 61 |
| ZC root index $u$ | 1 |
| Total bits evaluated | 100,000 |

The subcarrier spacing of 200 Hz and symbol duration of 20 ms are consistent ($200\,\text{Hz} = 1/20\,\text{ms}$), ensuring subcarrier orthogonality. The 5 ms cyclic prefix is designed to exceed the expected acoustic channel delay spread in a typical indoor room, preventing inter-symbol interference (ISI).

## III. Transmitter

### A. Bit Source

The binary data to be transmitted is loaded from a CSV file. The bit stream is zero-padded to the nearest multiple of 180 (one full OFDM symbol worth of bits), then reshaped into a matrix of shape $556 \times 180$, where each row corresponds to one OFDM symbol.

### B. Gray-Code Mapping

Each pair of bits $(b_0, b_1)$ is mapped to a differential phase increment $\Delta\phi$ according to a Gray-coded rule shown in Table II.

TABLE II
GRAY-CODED DQPSK PHASE MAP

| **Dibit** $(b_0, b_1)$ | **Phase Increment** $\Delta\phi$ |
|---|---|
| $(0, 0)$ | $0$ |
| $(0, 1)$ | $\pi/2$ |
| $(1, 1)$ | $\pi$ |
| $(1, 0)$ | $3\pi/2$ |

Gray coding ensures that adjacent constellation points differ by only one bit, so a single phase decision error causes only one bit error rather than two.

### C. Differential Encoding

Before any data symbols are sent, a reference OFDM symbol with all zeros (i.e., zero phase increment on every subcarrier) is prepended to the payload. This gives the receiver a known starting phase for the first differential step.

For each data symbol $k$ and subcarrier $m$, the absolute transmitted phase is accumulated:

$$\phi_k^{(m)} = \left(\phi_{k-1}^{(m)} + \Delta\phi_k^{(m)}\right) \bmod 2\pi, \tag{1}$$

and the complex OFDM subcarrier value is:

$$X_k[m] = e^{j\phi_k^{(m)}}. \tag{2}$$

### D. Subcarrier Mapping and Pilot Insertion

The 92 active subcarrier positions in the FFT frequency bin index are:

$$b_i = b_{\text{start}} + i \cdot \Delta b, \quad i = 0, 1, \ldots, 91, \tag{3}$$

where $b_{\text{start}} = \lfloor 1000/(F_s/N) \rfloor = 20$ and $\Delta b = \lfloor 200/(F_s/N) \rfloor = 2$, with $N = 960$ samples. Indices $i = 0$ and $i = 91$ carry pilot tones (constant amplitude unity); indices $i = 1$ through $i = 90$ carry the 90 DQPSK data symbols. To ensure a real-valued time-domain output (suitable for audio), the negative-frequency mirror bins are loaded with the complex conjugate:

$$X[-b] = X^*[b]. \tag{4}$$

### E. OFDM Modulation and Cyclic Prefix

The time-domain OFDM symbol is produced by the Inverse FFT:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k]\, e^{j2\pi kn/N}, \quad n = 0, \ldots, N-1. \tag{5}$$

The cyclic prefix is formed by copying the last $N_{\text{CP}} = 480$ samples of $x[n]$ to the front:

$$x_{\text{CP}}[n] = x[(n + N - N_{\text{CP}}) \bmod N], \quad n = 0, \ldots, N_{\text{CP}} - 1. \tag{6}$$

The full transmitted symbol is then $[x_{\text{CP}}; x]$ of length $N + N_{\text{CP}} = 1200$ samples.

### F. Windowing

A Tukey window ($\alpha = 0.05$) is applied to each CP-extended symbol before transmission:

$$x_w[n] = x_{\text{CP}}[n] \cdot w_{\text{Tukey}}[n]. \tag{7}$$

Windowing tapers the symbol boundaries, reducing spectral leakage and suppressing the audible click artifacts caused by abrupt amplitude transitions at symbol edges. With $\alpha = 0.05$ only 2.5% of each end of the symbol is tapered, so virtually no ISI is introduced.

### G. Zadoff–Chu Preamble

A Zadoff–Chu (ZC) sequence of length $N_{\text{ZC}} = 61$ with root index $u = 1$ is used as the synchronization and channel-estimation preamble:

$$z[n] = e^{-j\pi u\, n(n+1)/N_{\text{ZC}}}, \quad n = 0, \ldots, 60. \tag{8}$$

The ZC chips are placed into the same 61 lowest active frequency bins of an otherwise-zero FFT frame, and the IFFT produces the real-valued time-domain preamble waveform (with conjugate symmetry applied). The preamble is given its own CP and Tukey window, and is placed at the very start of the packet.

ZC sequences are chosen for this role because their periodic autocorrelation is a unit impulse: the cross-correlation of the received signal with a local copy produces a single sharp peak exactly at the start of the preamble, with zero sidelobes. This makes timing acquisition reliable even in noisy environments.

### H. Final Packet Assembly

The transmitted packet is assembled as:

$$[\, \underbrace{0 \cdots 0}_{1\text{ s silence}},\ \text{ZC preamble},\ \text{reference symbol},\ \underbrace{\text{data symbols}}_{556},\ \underbrace{0 \cdots 0}_{1\text{ s silence}} \,]$$

The leading and trailing silence prevents abrupt onset artifacts and gives the recording device time to settle.

The final waveform is peak-normalized to $[-1, +1]$ and written as 16-bit PCM audio at 48 kHz.

## IV. RECEIVER

### A. Audio Capture

The receiver records the acoustic channel output using the laptop microphone at 48 kHz. In addition to the transmitted waveform, the recording contains room reverberation, ambient noise, speaker and microphone non-linearities, and a small clock frequency mismatch between the two devices (CFO).

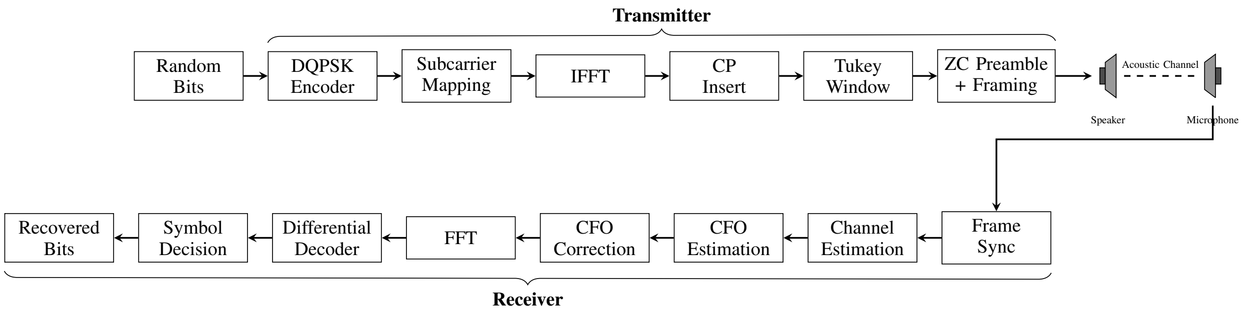


Fig. 1. Complete signal chain block diagram of the implemented system.

### B. Frame Synchronization

The receiver generates a local copy of the ZC preamble (identical construction to the transmitter) and computes a matched-filter correlation with the received signal using FFT-based convolution:

$$c[m] = \sum_n y[n] \cdot x^*_{\text{local}}[n-m]. \tag{9}$$

The index of the largest magnitude peak, $\hat{m}$, is taken as the start of the preamble:

$$\hat{m} = \arg\max_m |c[m]|. \tag{10}$$

The start of the data payload (reference symbol) is then:

$$n_{\text{ref}} = \hat{m} + (N + N_{\text{CP}}), \tag{11}$$

i.e., one full symbol length after the preamble start.

### C. Channel Estimation

After locating the preamble, the receiver strips the cyclic prefix and applies a power-normalized Tukey window ($\alpha = 0.1$) to the preamble body before taking the FFT:

$$Y_p[k] = \text{FFT}\big\{y_p[n] \cdot w_{\text{Tukey}}[n]\big\}. \tag{12}$$

A one-tap frequency-domain channel estimate is computed at each active ZC bin:

$$\hat{H}[k] = \frac{Y_p[k]}{X_p[k] + \varepsilon}, \quad \varepsilon = 10^{-12}, \tag{13}$$

where $X_p[k]$ is the known transmitted preamble spectrum. The channel estimate captures the complex gain (amplitude attenuation and phase shift) introduced by the acoustic path at each subcarrier frequency.

### D. Carrier Frequency Offset Estimation

A non-zero CFO causes every OFDM subcarrier to accumulate a linearly increasing phase rotation across successive symbols. The pilot tones — which are transmitted as unit-amplitude real values and are therefore at a known phase each symbol — serve as a clean reference for measuring this drift.

In a first pass over all symbols, the receiver extracts the raw phase of both pilot tones without any windowing (the rectangular FFT window is the matched filter for a pure sinusoid):

$$\theta_k^{(p)} = \angle X_{\text{rx}}^{(k)}[b_p], \quad k = 0, \ldots, K-1. \tag{14}$$

Each phase sequence is unwrapped to remove $2\pi$ discontinuities, and a linear least-squares fit is applied to the pilot-B sequence:

$$\hat{\theta}_k = a\,k + b, \quad \hat{a}, \hat{b} = \arg\min \sum_k (\theta_k^{(B)} - ak - b)^2. \tag{15}$$

The estimated CFO in Hz is:

$$\hat{f}_{\text{CFO}} = \frac{\hat{a}}{2\pi\, T_{\text{sym}}}. \tag{16}$$

In the recorded experiment, the estimated CFO was $-0.147$ Hz, corresponding to a phase drift of $-0.0185$ rad per symbol — small but non-negligible over a 556-symbol packet.

### E. Symbol Extraction and CFO Correction

In a second pass, each data symbol is extracted and CFO-corrected before demodulation. After stripping the cyclic prefix, the predicted bulk phase accumulated by symbol $k$ is:

$$\hat{\psi}_k = \hat{a} \cdot k. \tag{17}$$

This is applied as a frequency-domain rotation:

$$\mathbf{X}_{\text{corr}}^{(k)} = \text{FFT}\{y_{\text{core}}^{(k)}\} \cdot e^{-j\hat{\psi}_k}, \tag{18}$$

which is equivalent to compensating the inter-symbol phase accumulation caused by the CFO. Note that this approach corrects the *cumulative* phase drift per symbol; it does not correct the *within-symbol* phase ramp, which is small for sub-Hz offsets on a 20 ms symbol.

No equalization by $\hat{H}$ is applied to the data symbols in this receiver; the differential decoding step (Section IV-F) inherently cancels any slowly-varying channel phase and amplitude eq. was unnecessary because DQPSK detection is phase-based.

### F. Differential Decoding

Let $s_k^{(m)}$ denote the complex symbol extracted at subcarrier $m$ in symbol $k$ after CFO correction. Differential decoding is performed by conjugate multiplication of consecutive symbols:

$$d_k^{(m)} = s_k^{(m)} \cdot \left(s_{k-1}^{(m)}\right)^*, \tag{19}$$

The angle of $d_k^{(m)}$ is the recovered differential phase:

$$\hat{\Delta\phi}_k^{(m)} = \angle\, d_k^{(m)} \bmod 2\pi. \tag{20}$$

### G. Symbol Decision and Bit Recovery

The decoded phase is compared against the four candidate phases $\{0, \pi/2, \pi, 3\pi/2\}$ using circular nearest-neighbour detection:

$$\hat{i}_k^{(m)} = \arg\min_i \, d_{\text{circ}}\left(\hat{\Delta\phi}_k^{(m)},\ \phi_i\right), \tag{21}$$

where $d_{\text{circ}}(\alpha, \beta) = \min(|\alpha - \beta|,\ 2\pi - |\alpha - \beta|)$ is the circular distance. The winning index $\hat{i}$ is then mapped back to a dibit $(b_0, b_1)$ using the inverse of Table II.

## V. Results and Analysis

### A. System Performance

The transceiver was evaluated on a live acoustic recording captured with a laptop microphone in an indoor environment. Table III summarizes the measured performance.

TABLE III
MEASURED SYSTEM PERFORMANCE

| **Metric** | **Value** |
|---|---|
| Total bits evaluated | 100,000 |
| Bit errors | 1,286 |
| BER | $\approx 1.3\%$ |
| Gross data rate | $\approx 7.2$ kbps |
| Estimated CFO | $-0.147$ Hz |
| CFO phase drift | $-0.0185$ rad/symbol |

In a practical system, a forward error correction (FEC) code such as a rate-1/2 convolutional code or a Reed–Solomon code would be added on top to bring the post-decoding BER down even further.

### B. DQPSK Constellation

Figure 2 shows the truth-colored differential constellation: each received point $d_k^{(m)}$ is plotted in the complex plane and colored according to the *true* transmitted dibit. Points that appear in the wrong color quadrant represent bit errors.

The constellation shows four distinct clusters aligned with the four ideal phase axes $(1+0j,\ 0+1j,\ -1+0j,\ 0-1j)$, confirming that the system is correctly recovering differential phase. The radial spread within each cluster (i.e., the varying distance from the origin) reflects amplitude modulation by the acoustic channel, which is expected: the differential decoder cares only about the angle of $d_k^{(m)}$, not its magnitude. The angular spread within each cluster is the primary source of BER, driven by additive noise and residual ISI.

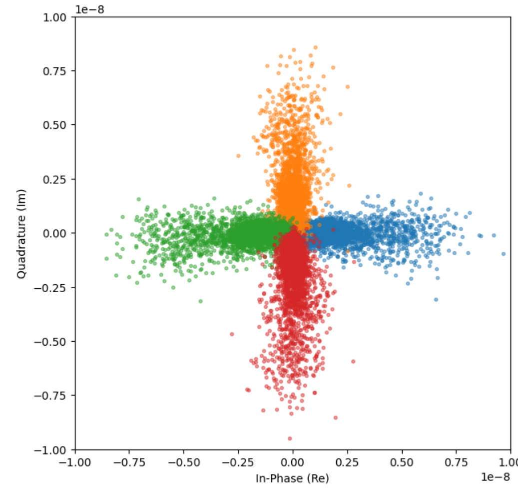


Fig. 2. DQPSK differential constellation. Each point is $d_k^{(m)} = s_k^{(m)} \cdot (s_{k-1}^{(m)})^*$.

### C. Differential Phase Histogram

Figure 3 shows the histogram of recovered differential phases $\hat{\Delta\phi}$ across all symbols and subcarriers. Four dominant peaks are visible near $0$, $\pi/2$, $\pi$, and $3\pi/2$, confirming successful differential decoding. The finite width of each peak reflects the combined effect of channel noise, residual ISI from imperfect CP coverage, and the small residual CFO after correction. The roughly equal heights of the four peaks are consistent with the approximately uniform distribution expected from a random bit source.

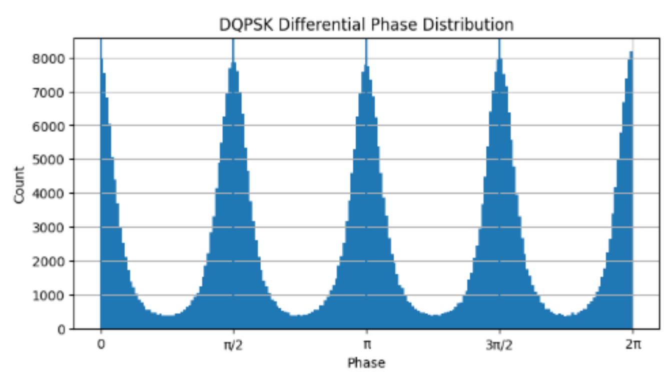


Fig. 3. Histogram of recovered DQPSK differential phases. Vertical dashed lines mark the four ideal phase locations.

### D. Time Synchronization

The matched-filter correlation output produced a single dominant peak at sample 70,768, corresponding to a payload start time of approximately 1.50 s into the recording (including the 1 s of leading silence). The sharpness of the ZC correlation peak allows reliable synchronization even in the presence of room echoes, which appear as lower secondary peaks in the correlation output.

### E. CFO Estimation

The unwrapped pilot-B phase sequence displayed a consistent linear trend across all 557 symbols, with a fitted slope of $-0.0185$ rad per symbol ($\hat{f}_{\text{CFO}} = -0.147$ Hz). The small but consistent CFO is attributable to the slight clock rate mismatch between the transmitting and receiving devices. Without correction, this drift would accumulate to $-10.3$ rad by the last symbol, which is more than one full revolution and sufficient to cause widespread decision errors. The linear-fit CFO correction successfully removed this drift.

## VI. Limitations

The system has several limitations that constrain its performance.

**Quasi-static channel assumption.** The preamble is at the start of the packet. The channel estimate is valid only if the channel does not change significantly over the 14 s packet duration. In a typical indoor room with stationary hardware this holds approximately, but any movement during transmission degrades performance.

**Within-symbol CFO phase ramp.** The CFO correction removes the per-symbol bulk phase, but does not correct the phase ramp within each 20 ms symbol. At 0.147 Hz this intra-symbol phase error is negligible ($< 0.002$ rad over one symbol), but would become significant at larger CFO values.

**Sensitivity to parameter tuning.** More sophisticated receivers incorporating Schmidl–Cox synchronization, fine CFO correction, and comb-pilot interpolation-based equalization were explored during development but consistently proved sensitive to acoustic environment conditions, often failing catastrophically (BER $\approx 0.5$) when parameters were not precisely tuned. The simpler two-pass design described here was ultimately selected for its robustness.

## VII. Conclusion

A complete OFDM-DQPSK audio transceiver was implemented in Python and evaluated over a real acoustic channel. The system achieved a BER of 1.3% at a gross data rate of 7.2 kbps using 90 subcarriers across a 1–19.4 kHz audio band, with a 5 ms cyclic prefix for multipath robustness. Key design decisions — differential modulation for phase-error tolerance, ZC preamble for reliable timing and channel estimation, and pilot-based CFO tracking — all contributed to reliable operation on commodity hardware without any custom signal processing components. Future extensions could focus on applying the preamble channel estimate to the data symbols for per-subcarrier equalization, FEC to reduce the BER further.